

[illegible][illegible]

Pinout diagram for ESP32-53-WROOM-10U-N16R2. The diagram shows the pin numbers (1-16) on the left and the corresponding pin names on the right. The pins are arranged in a single row, with the GND symbol at the bottom and the VCC symbol at the top.

Pin Number	Pin Name
1	EN
2	IO0
3	INT1
4	IO2
5	IO3
6	INT1
7	IO4
8	IO5
9	IO6
10	IO7
11	IO8
12	IO9
13	IO10
14	IO11
15	IO12
16	IO13
17	IO14
18	IO15
19	IO16
20	IO17
21	IO18
22	IO19
23	IO20
24	IO21
25	IO22
26	IO23
27	IO24
28	IO25
29	IO26
30	IO27
31	IO28
32	IO29
33	IO30
34	IO31
35	IO32
36	IO33
37	IO34
38	IO35
39	IO36
40	IO37
41	IO38
42	IO39
43	IO40
44	IO41
45	IO42
46	IO43
47	IO44
48	IO45
49	IO46
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57	IO54
58	IO55
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63	IO60
64	IO61
65	IO62
66	IO63
67	IO64
68	IO65
69	IO66
70	IO67
71	IO68
72	IO69
73	IO70
74	IO71
75	IO72
76	IO73
77	IO74
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79	IO76
80	IO77
81	IO78
82	IO79
83	IO80
84	IO81
85	IO82
86	IO83
87	IO84
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89	IO86
90	IO87
91	IO88
92	IO89
93	IO90
94	IO91
95	IO92
96	IO93
97	IO94
98	IO95
99	IO96
100	IO97
101	IO98
102	IO99
103	IO100
104	IO101
105	IO102
106	IO103
107	IO104
108	IO105
109	IO106
110	IO107
111	IO108
112	IO109
113	IO110
114	IO111
115	IO112
116	IO113
117	IO114
118	IO115
119	IO116
120	IO117
121	IO118
122	IO119
123	IO120
124	IO121
125	IO122
126	IO123
127	IO124
128	IO125
129	IO126
130	IO127
131	IO128
132	IO129
133	IO130
134	IO131
135	IO132
136	IO133
137	IO134
138	IO135
139	IO136
140	IO137
141	IO138
142	IO139
143	IO140
144	IO141
145	IO142
146	IO143
147	IO144
148	IO145
149	IO146
150	IO147
151	IO148
152	IO149
153	IO150
154	IO151
155	IO152
156	IO153
157	IO154
158	IO155
159	IO156
1	

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SSE terminals iN9/30 pin = VINLP (4 Vdc, fused and VIO-clamped).
ALWAYS LINES! pin = AINCHV-LINE (fused and VIO-clamped). The watchdog gates
the shared return (Q7), NOT the supply, so a flipped watchdog
leaves 24 V on pin 1 and opens pin 2. Boot-stops R7/R20 hold
each gate down while the optics are done and the CPU boots.
The optics are NOT an isolation barrier – both sides share GND.
ISO-RTS device was designed as a 16-bit parallel bus.
UP/US or ISO is kept at 24 V gate float after the GPIO.
Per channel ~+20 mV, 1 bit holds at 750 nA and takes ~250
mA. I/O pins have 100 pF capacitors and take ~1 mA @ 24V.
T1 terminal J15 / T12 terminal J8B pin# = Kx, pin# = k – both
float. Blasted near GND terminal through MAX31865's internal
BAS network. Ungrounded conductors produce noise and
ground-junction problems in one kin could walk through it.
Near switch J16 is panel-mounted; inputs ESPR2 pull-ups.
J11 IN1/IN2/IN3 (1Id/gas-flow/switch) + GND – dry contact closed
terminal to GND.
Display J15 (pin #17)=P7H965 + XP72046 two-channel LCDWIKI
PWR ADDRESS 1=+5V 2=GND 3=C/LX 4=SPST 5=GND 6=PWR 7=OSC 8=B/L
9=PWR 10=GND 11=C/LX 14=LX 15=LX 16=LX 17=LX 18=LX 19=LX 20=LX
21=SPST 22=+5V 23=NOT SPST 24=NOT SPST 25=NOT SPST 26=NOT
wire a CXY-only panel to it. Logic is 3.3V, so shares SPIZ
with the LCD and both MAX31865s (R93-A3 damping).
T12 CT current inputs, CTA/CPT/CTB/CDB/P/CTBL... – one CT clamp
per SSR zone.
J15 DNP SPST voltage sense header, PINS -mated, rated NOT
to do wire to AC mains. V1/C25/C26 load caps are an ASSUMED
usage (unverified C.L.). Do not electrolytic connect. AD9593 IC2
address 0x3C collides with PCBF85A1.
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The schematic diagram illustrates the multi-channel system architecture. It features two horizontal rows of terminal pairs (TP) and associated processing blocks. The top row contains TP1 through TP7, while the bottom row contains TP8 through TP12. Vertical lines connect corresponding TPs between the two rows, indicating signal flow. Key components include several 'SP' blocks (e.g., SP1-WDS1, SP1-ESLCA, SP1-WDS0) which likely represent sensor or processing units, and a 'CTA-P' block. A power supply section at the top left shows connections for +3V3 and +5V, with TP3 TP grounded.